Meaning and characteristics of action research

Action research is a research method that aims to simultaneously **investigate and solve an issue**. In other words, as its name suggests, action research conducts research and takes action at the same time. It was first coined as a term in 1944 by MIT professor Kurt Lewin.A highly interactive method, action research is often used in the social sciences, particularly in educational settings. Particularly popular with educators as a form of systematic inquiry, it prioritizes reflection and bridges the gap between theory and practice. Due to the nature of the research, it is also sometimes called a **cycle of action** or a **cycle of inquiry**.

Types of action research

There are 2 common types of action research: participatory action research and practical action research.

- **Participatory action research** emphasizes that participants should be members of the community being studied, empowering those directly affected by outcomes of said research. In this method, participants are effectively co-researchers, with their lived experiences considered formative to the research process.
- Example: Addressing Food Insecurity in a Local Community
- In this scenario, researchers collaborate with members of a low-income neighborhood facing food insecurity issues. The researchers and community members work together to identify the root causes of food insecurity, such as lack of access to affordable groceries or knowledge about nutrition. Through a series of meetings, surveys, and focus groups, the researchers gather data and insights from community members, who actively participate in the research process. Together, they develop and implement strategies to address these issues, such as establishing community gardens, organizing cooking workshops, or advocating for policy changes. Throughout the process, community members are empowered to take ownership of the solutions, building their capacity to address similar challenges in the future.

- **Practical action research** focuses more on how research is conducted and is designed to address and solve specific issues.
- Example: Improving Classroom Learning Outcomes
- In this example, a teacher conducts action research to enhance student engagement and learning outcomes in their classroom. The teacher identifies a specific challenge, such as students' difficulty understanding mathematical concepts. They then design and implement interventions, such as incorporating hands-on activities, collaborative learning techniques, or technology-enhanced lessons. The teacher collects data on student performance, behavior, and feedback before and after implementing the interventions. Through reflection and analysis of the data, the teacher evaluates the effectiveness of each intervention and makes adjustments as needed. By actively engaging in this research process, the teacher not only improves learning outcomes in their own classroom but also gains valuable insights and skills that can be applied in future teaching practice.

- **Examples of action research**

- Action research is often used in fields like education because of its iterative and flexible style.
- Example: Participatory action research : As part of an ongoing commitment to improve school facilities for students with disabilities, an action research plan asked students using wheelchairs to time how long it took them to get to and from various points on school grounds.
- After the information was collected, the students were asked where they thought ramps or other accessibility measures would be best utilized, and the suggestions were sent to school administrators.
- Example: Practical action research Science teachers at your city's high school have been witnessing a year-over-year decline in standardized test scores in chemistry. In seeking the source of this issue, they studied how concepts are taught in depth, focusing on the methods, tools, and approaches used by each teacher.
- They found that there had been no change in how chemistry was taught in the last decade—with no incorporation of more modernized teaching approaches or useful online tools. Teachers resolved to implement more modern techniques in their teaching to see if that could improve scores.

Both types of action research are more focused on increasing the capacity and ability of future practitioners than contributing to a theoretical body of knowledge.

he main characteristics of action research include:

1. **Participation and Collaboration**: Action research involves active participation and collaboration between researchers and stakeholders, such as community members, practitioners, or organizational members. It emphasizes the involvement of those directly affected by the research problem in all stages of the research process.
2. **Cyclical Process**: Action research typically follows a cyclical process of planning, action, observation, and reflection. Researchers engage in iterative cycles of action and reflection to continuously refine and improve their understanding of the problem and develop effective solutions.
3. **Empowerment and Capacity Building**: Action research aims to empower participants by involving them as co-researchers in the research process. It seeks to build the capacity of individuals and communities to identify and address their own challenges, rather than relying solely on external experts or researchers.
4. **Practical Orientation**: Action research is oriented towards addressing practical problems and generating actionable knowledge that can lead to tangible improvements in real-world situations. It is often conducted in applied settings, such as classrooms, organizations, or communities, with the goal of producing solutions that are relevant and useful to stakeholders.
5. **Reflective Inquiry**: Reflection is a key component of action research, allowing researchers and participants to critically examine their assumptions, beliefs, and practices. Through ongoing reflection, researchers can deepen their understanding of the research problem, evaluate the effectiveness of interventions, and identify areas for further inquiry and improvement.

6. **Contextual Sensitivity**: Action research recognizes the importance of context in shaping research questions, methods, and findings. It takes into account the unique characteristics, values, and dynamics of the settings in which it is conducted, ensuring that interventions are tailored to the specific needs and realities of participants.
7. **Iterative Learning and Adaptation**: Action research is characterized by a willingness to experiment, learn from failures, and adapt strategies based on feedback and evidence. Researchers are open to revising their approaches in response to new insights or changing circumstances, with the goal of continuously improving outcomes over time.

Action research vs. traditional research

Action research differs sharply from other types of research in that it seeks to produce actionable processes over the course of the research rather than contributing to existing knowledge or drawing conclusions from datasets. In this way, action research is **formative**, not **summative**, and is conducted in an ongoing, iterative way.

	Action research	**Traditional research**
Purpose	• Solve immediate problems • Improve existing systems	• Draw conclusions from data • Advance existing knowledge • Provide generalizable and reliable findings
Context	• Reactive, derived from surroundings • Usually not theoretical in nature	• Focused on crafting strong hypotheses and seeking causal relationships between variables • Derived from theory
Significance	• Practical	• Statistical

As such, action research is different in purpose, context, and significance and is a good fit for those seeking to implement systemic change.

Advantages and disadvantages of action research

Action research comes with advantages and disadvantages.

Advantages

- Action research is **highly adaptable**, allowing researchers to mold their analysis to their individual needs and implement practical individual-level changes.
- Action research provides an **immediate and actionable path** forward for solving entrenched issues, rather than suggesting complicated, longer-term solutions rooted in complex data.
- Done correctly, action research can be very **empowering**, informing social change and allowing participants to effect that change in ways meaningful to their communities.

Disadvantages

- Due to their flexibility, action research studies are plagued by very **limited generalizability** and are very **difficult to replicate**. They are often not considered theoretically rigorous due to the power the researcher holds in drawing conclusions.
- Action research can be **complicated to structure in an ethical manner**. Participants may feel pressured to participate or to participate in a certain way.
- Action research is at **high risk for research biases** such as selection bias, social desirability bias, or other types of cognitive biases.

4.1 Action research cycle

4.1.1 Planning
4.1.2 Acting
4.1.3 Observing
4.1.4 Reflecting

he action research cycle typically involves four main stages: Planning, Acting, Observing, and Reflecting. Here's an example of the action research cycle in the context of improving reading comprehension in a primary school classroom:

1. **Planning**:
    - Identify the Problem: The teacher notices that several students in their Grade 3 class are struggling with reading comprehension.
    - Set Goals: The teacher aims to improve students' reading comprehension skills by implementing targeted interventions.
    - Develop Action Plan: The teacher plans to incorporate reading comprehension strategies, such as predicting, summarizing, and questioning, into daily reading activities.
2. **Acting**:
    - Implement Interventions: The teacher introduces the selected reading comprehension strategies during reading sessions. For example, they model how to make predictions before reading a text and encourage students to ask questions while reading.
    - Provide Support: The teacher offers individualized support to students who need extra help, such as providing additional practice materials or arranging peer tutoring sessions.
3. **Observing**:
    - Collect Data: The teacher observes students' engagement and comprehension during reading activities, takes anecdotal notes, and administers pre- and post-assessments to measure progress.
    - Analyze Data: The teacher analyzes the collected data to identify trends, patterns, and areas of improvement. They compare pre- and post-assessment scores to assess the effectiveness of the interventions.
4. **Reflecting**:
    - Reflect on Results: Based on the data analysis, the teacher reflects on what worked well and what could be improved in the interventions. They consider factors such as student engagement, comprehension levels, and the effectiveness of different strategies.
    - Adjust Strategies: The teacher modifies the action plan based on their reflections, incorporating feedback from students and colleagues. They may refine existing strategies, introduce new interventions, or address any challenges that arose during implementation.
    - Plan Next Steps: The teacher develops a plan for the next cycle of action research, setting new goals and strategies for further improvement. They may also share their findings and insights with colleagues to inform teaching practices across the school.

By following the action research cycle, the teacher engages in a systematic process of inquiry and improvement, continuously refining their teaching practices to better support students' learning needs. Through reflection and collaboration, they contribute to ongoing professional development and the advancement of effective instructional strategies.

4.2 Preparing action research report